

Hriday Sainathuni

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EDUCATION

University of Virginia, School of Engineering and Applied Science

Charlottesville, VA

Intended Major: B.S. Computer Science

Expected Graduation: May 2027

Relevant Coursework: Data Structures & Algorithms II, Computer Systems I-II, Discrete Math, Software Development, Cybersecurity

Thomas Jefferson High School for Science and Technology - Prestigious STEM-focused high school

Graduated 2024

EXPERIENCE

Software Engineering Intern | NuAxis (Solutions Engineering Group)

Aug 2025 - Current

- Prototyped **LLM-powered solutions on Azure** to improve federal digital services and streamline **citizen permit applications**
- Developed **chatbot workflows** using **OpenAI APIs**, **LangChain**, and **PostgreSQL** for automated permit processing
- Built **full-stack AI prototypes** with **React**, **MongoDB**, and **Postgres**, collaborating in **Agile sprints** on **Git workflows** and rapid prototyping of scalable government tech solutions
- Conducted **content and accessibility analysis** to optimize digital customer experience (**CX**) across federal websites

AI/ML Intern | Verizon (Cyber AI Services Team)

May 2025 - Aug 2025

- Built a **Dockerized FastAPI REST platform** for speech-to-speech GenAI evaluation using **Twilio** and **agentic LLMs**
- Integrated **Vertex AI** into CI/CD pipelines with **MLOps workflows** to automate GenAI vulnerability detection
- Automated **adversarial security evaluations** across speech modalities, including prompt injection and robustness testing
- Developed a **deepfake synthesis microservice** (Chatterbox-TTS, LatentSync) deployed on **GCP**; presented outputs to Verizon's phishing team and executives to raise threat awareness

Undergraduate Research Assistant | CRDL Lab at the University of Virginia

Aug 2024 - Aug 2025

- **Co-authored** an **ICRA 2025** paper on decentralized multi-agent coordination in robotic navigation
- Built **SMGLib**, a simulation library using **Gym**, **PyTorch**, and **TensorFlow** for path planning benchmarks
- Trained **PPO reinforcement learning policies** and applied optimization for deadlock resolution
- Produced **trajectory datasets**, **evaluation metrics**, and **visualizations** with OpenCV and Matplotlib

Aspiring Scientists Summer Internship Program | George Mason University

May 2023 - Jun 2024

- Simulated **collaborative robot behavior** using NetworkX and interactive visualization tools (Dash, Plotly)
- Implemented and tested **RL algorithms** (Q-Learning, PPO, hierarchical decomposition) in dynamic multi-agent environments
- Created evaluation figures and experimental results for an **IEEE ICRA submission** analyzing learned policy generalization

Research Intern | Virginia Tech Sanghani Center

Nov 2022 - May 2024

- Developed a GenAI image classification pipeline using subclass labels from COCO and Conceptual Captions to probe features
- Designed **prompting strategies** to reveal **LLM reasoning**, visualized with **BERT-based** tool for interpretability
- Credited in a **CVPR 2024 publication** on black-box analysis of vision-language models and cross-modal reasoning

PROJECTS

UVA Career Center Chatbot | University of Virginia

May 2025 - Current

- Developing a UVA Career Center chatbot using **Google Gemini** with a **RAG pipeline** powered by **ChromaDB**
- Building a **React/Next.js** front-end deployed on a dedicated **sub-domain** and hosted on **AWS** for scalable, reliable delivery

Entity Linking for Multimodal Knowledge Bases | Virginia Tech NLP Lab

Nov 2023 - Feb 2024

- Built a **benchmark test set** for evaluating **cross-modal reasoning** in entity linking tasks across text, image, and KB metadata
- Developed preprocessing pipelines to align multimodal content for evaluating **RAG pipelines** and vision-language models
- Analyzed model **failure cases** in multimodal inference and contributed evaluation feedback to internal model iteration loops

Multimodal Data Platform Development | RobotiXX Lab, George Mason University

Jun 2023 - Aug 2023

- Engineered a ROS-based mobile device to collect multimodal sensor data (LiDAR, audio, video, odometry) for SLAM research
- Automated sensor calibration and deployment via ROS launch configs; dataset was used in an **IROS 2023 publication**

SKILLS

Languages & Libraries: Python, C++, Java, SQL, JavaScript, PyTorch, TensorFlow, Keras, HuggingFace, OpenCV, scikit-learn

Tools & Platforms: Docker, GCP, AWS, Azure, Git, ROS, Burp Suite, NetworkX, Dash, FastAPI, ChromaDB, Vertex AI